Latex Formular Figure Table OCR Sample

Mark Yu

September 2026

1 Formular

1.1 Single column

Single-line formula:

$$a^2 + b^2 = c^2$$

Numbered formula:

$$E = mc^2 \tag{1}$$

Complex summation:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} x_j = b_i \tag{2}$$

Multiple integral:

$$\iint_D f(x, y)\, dx\, dy \tag{3}$$

Limit and fraction:

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e \tag{4}$$

Matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{5}$$

Piecewise function:

$$f(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0. \end{cases} \tag{6}$$

Multi-line formula:

$$y = ax^2 + bx + c \tag{7}$$

$$\Delta = b^2 - 4ac \tag{8}$$

1.2 Double column

Summation formula:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n \tag{9}$$

Integral formula:

$$\int_a^b f(x)\,dx = F(b) - F(a) \tag{10}$$

Limit formula:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \tag{11}$$

Aligned multi-line formula:

$$(a+b)^2 = a^2 + 2ab + b^2 \tag{12}$$
$$(a-b)^2 = a^2 - 2ab + b^2 \tag{13}$$

Complex summation:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} x_j = b_i \tag{14}$$

Multiple integral:

$$\iint_D f(x, y)\,dx\,dy \tag{15}$$

Limit and fraction:

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e \tag{16}$$

Matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{17}$$

Piecewise function:

$$f(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0. \end{cases} \tag{18}$$

Multi-line formula:

$$y = ax^2 + bx + c \tag{19}$$
$$\Delta = b^2 - 4ac \tag{20}$$

1.3 Inline Formulas

In mathematics, the quadratic equation $ax^2 + bx + c = 0$ can be solved using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where the discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots. When $\Delta > 0$, there are two distinct real roots; when $\Delta = 0$, there is exactly one real root; and when $\Delta < 0$, the roots are complex conjugates.

For sequences, the arithmetic series $\sum_{i=1}^{n} a_i$ can be expressed as $\frac{n(a_1 + a_n)}{2}$, while the geometric series $\sum_{k=0}^{\infty} ar^k$ converges to $\frac{a}{1-r}$ when $|r| < 1$. In calculus, the derivative $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ represents the instantaneous rate of change, and the definite integral $\int_a^b f(x)\,dx$ gives the accumulated area under the curve.

In probability theory, if $X \sim N(\mu, \sigma^2)$, then the expected value $\mathbb{E}[X] = \mu$ and the variance $\mathrm{Var}(X) = \sigma^2$. The standard normal distribution has the density function $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and its cumulative distribution function is denoted by $\Phi(x)$.

For vectors in $\mathbb{R}^n$, the dot product $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$ measures the angle between vectors, while the norm $\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{n} v_i^2}$ gives the Euclidean length. A matrix $A \in \mathbb{R}^{m \times n}$ is orthogonal if $A^T A = I_n$, where I_n is the $n \times n$ identity matrix.

In information theory, the entropy $H(X) = -\sum_i p(x_i) \log p(x_i)$ quantifies

uncertainty, and the KL divergence $D_{\mathrm{KL}}(P\|Q) = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$ measures the difference between two probability distributions P and Q.

In mathematics, the quadratic equation $ax^2+bx+c=0$ can be solved using the formula $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$, where the discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots. When $\Delta > 0$, there are two distinct real roots; when $\Delta = 0$, there is exactly one real root; and when $\Delta < 0$, the roots are complex conjugates.

For sequences, the arithmetic series $\sum_{i=1}^{n} a_i$ can be expressed as $\frac{n(a_1+a_n)}{2}$, while the geometric series $\sum_{k=0}^{\infty} ar^k$ converges to $\frac{a}{1-r}$ when $|r| < 1$. In calculus, the derivative $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ represents the instantaneous rate of change, and the definite integral $\int_a^b f(x)\,dx$ gives the accumulated area under the curve.

In probability theory, if $X \sim N(\mu, \sigma^2)$, then the expected value $\mathbb{E}[X] = \mu$ and the variance $\mathrm{Var}(X) = \sigma^2$. The standard normal distribution has the density function $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and its cumulative distribution function is denoted by $\Phi(x)$.

For vectors in $\mathbb{R}^n$, the dot product $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$ measures the angle between vectors, while the norm $\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{n} v_i^2}$ gives the Euclidean length. A matrix $A \in \mathbb{R}^{m\times n}$ is orthogonal if $A^T A = I_n$, where I_n is the $n \times n$ identity matrix.

In information theory, the entropy $H(X) = -\sum_i p(x_i) \log p(x_i)$ quantifies uncertainty, and the KL divergence $D_{\mathrm{KL}}(P\|Q) = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$ measures the difference between two probability distributions P and Q.

2 Figure

2.1 Single column

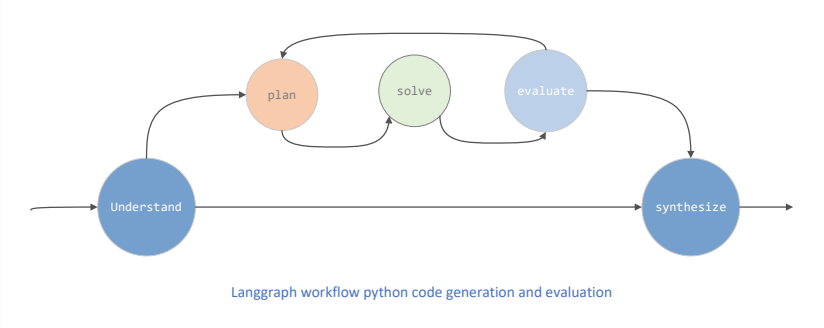
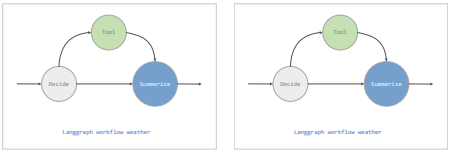


Figure 1: Single column figure sample.

2.2 Double column



(a) First subfigure *(b) Second subfigure*

Figure 1: Main caption

3 Table

3.1 Single column

Table 1: Single Column Table Example

Method	Accuracy	F1 Score
Baseline	0.85	0.83
Proposed	0.92	0.91

3.2 Double column

Double column table with adjusted width:

Table 2: Double Column Table Example

Method	Acc.	F1
SVM	0.82	0.80
RF	0.87	0.86
CNN	0.90	0.89

Another compact table:

Table 3: Compact Table

Item	Value
A	1.23
B	4.56
C	7.89

4 Reference and citation

4.1 Numeric citation style

This is a citation [1].

Multiple citations can be combined [2, 3], or cited with specific context [see 4, Chapter 3].

According to **(author?)** [1], the self-attention mechanism has revolutionized natural language processing. Recent studies [2, 3] further advanced the field.

4.2 Author-year citation style

The transformer architecture [1] revolutionized NLP. **(author?)** [3] further advanced pre-training techniques.

Combined citations [2, 3] show the progression. Textual citation: **(author?)** [4] provides comprehensive coverage.

References

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30:5998–6008, 2017.

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

[4] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.